

A Polynomial-Time Approximation Scheme for Expected Average Nash Welfare under Interim Envy-Freeness

Draft note

May 29, 2026

Abstract

Caragiannis, Kanellopoulos, and Kyropoulou gave polynomial-time algorithms for computing interim envy-free lotteries that maximize expected utilitarian, egalitarian, and log-Nash welfare in matching allocation instances. They explicitly left open the exact computation of an interim envy-free lottery of maximum expected average Nash welfare. This note observes that their linear-programming and separation-oracle framework yields a multiplicative approximation scheme for that objective. For every $\varepsilon \in (0, 1)$, one can compute an interim envy-free lottery whose expected average Nash welfare is at least $(1 - \varepsilon)$ times optimal, with polynomial support and polynomially many calls to the same edge-pair-weighted matching oracle used for the log-Nash objective. The argument replaces the exponential map relating log-Nash and average Nash welfare by a finite lower envelope of tangent lines. The result is an approximation theorem; it does not resolve the exact computation problem.

1 Introduction

A matching allocation instance consists of a set N of n agents, a set I of n items, and nonnegative valuations $v_i(g)$ for each agent $i \in N$ and item $g \in I$. An allocation is a perfect matching $\mu : N \rightarrow I$; agent i receives item $\mu(i)$. A lottery is a probability distribution over perfect matchings.

Caragiannis, Kanellopoulos, and Kyropoulou introduced and studied interim envy-freeness, abbreviated iEF, for lotteries over allocations [1]. In matching allocation instances, they show how to compute, in polynomial time, iEF lotteries that maximize expected utilitarian, egalitarian, and log-Nash welfare. Their method formulates the problem as a linear program with one variable per perfect matching, solves the dual by the ellipsoid method, and implements separation through a polynomial-time algorithm for maximum edge-pair-weighted bipartite perfect matching, called 2EBM. They also point out that maximizing expected average Nash welfare is not equivalent to maximizing expected log-Nash welfare for lotteries, and leave exact computation of maximum expected average Nash welfare open.

This note records a simple consequence of the same machinery. Let

$$\text{aNW}(\mu) = \left(\prod_{i \in N} v_i(\mu(i)) \right)^{1/n}$$

denote the average Nash welfare of matching μ . The main observation is that, on positive-welfare matchings,

$$\text{aNW}(\mu) = \exp \left(\frac{1}{n} \sum_{i \in N} \log v_i(\mu(i)) \right).$$

The log-Nash term inside the exponential is additive over the chosen edges of the matching. A finite tangent approximation from below to the exponential therefore converts average Nash welfare into the maximum of polynomially many log-Nash-type additive objectives. The dual separation problem for the approximating LP reduces to polynomially many 2EBM calls.

The approximation theorem is stated in the same real-arithmetic/separation-oracle model as the log-Nash computation of [1]. A fully bit-complexity implementation requires the standard additional bookkeeping for approximating logarithmic and exponential coefficients; this is orthogonal to the combinatorial point of the note.

2 Preliminaries

Let $\mathcal{M}$ denote the set of perfect matchings between N and I . For $i, k \in N$, $g \in I$, and $\mu \in \mathcal{M}$, define

$$A_{i,k,g}(\mu) = \mathbf{1}[\mu(i) = g](v_i(g) - v_i(\mu(k))).$$

A lottery $q = (q_\mu)_{\mu \in \mathcal{M}}$ is iEF exactly when

$$\sum_{\mu \in \mathcal{M}} q_\mu A_{i,k,g}(\mu) \geq 0 \quad \text{for every } i, k \in N, g \in I, \quad (1)$$

together with

$$\sum_{\mu \in \mathcal{M}} q_\mu = 1, \quad q_\mu \geq 0 \quad (\mu \in \mathcal{M}).$$

Indeed, when agent i receives item g with positive probability, (1) is exactly the iEF condition

$$v_i(g) \geq \mathbb{E}_{\mu \sim q} [v_i(\mu(k)) \mid \mu(i) = g],$$

multiplied by $\Pr_{\mu \sim q}[\mu(i) = g]$. When this probability is zero, the corresponding inequality is vacuous.

For any coefficient vector $c = (c_\mu)_{\mu \in \mathcal{M}}$, consider the welfare-maximization LP

$$\begin{aligned} \max \quad & \sum_{\mu \in \mathcal{M}} q_\mu c_\mu \\ \text{s.t.} \quad & \sum_{\mu \in \mathcal{M}} q_\mu A_{i,k,g}(\mu) \geq 0 \quad \forall i, k \in N, g \in I, \\ & \sum_{\mu \in \mathcal{M}} q_\mu = 1, \\ & q_\mu \geq 0 \quad \forall \mu \in \mathcal{M}. \end{aligned} \quad (2)$$

The exact expected average-Nash problem is obtained by taking $c_\mu = \text{aNW}(\mu)$.

The dual of (2) has nonnegative variables $y_{i,k,g}$ for the iEF constraints and a free variable z for the probability constraint. It is

$$\begin{aligned} \min \quad & z \\ \text{s.t.} \quad & z \geq c_\mu + B_y(\mu) \quad \forall \mu \in \mathcal{M}, \\ & y_{i,k,g} \geq 0 \quad \forall i, k \in N, g \in I, \end{aligned} \quad (3)$$

where

$$B_y(\mu) = \sum_{i,k \in N} \sum_{g \in I} y_{i,k,g} A_{i,k,g}(\mu). \quad (4)$$

Thus a separation oracle for the dual must maximize

$$B_y(\mu) + c_\mu \tag{5}$$

over perfect matchings μ .

The term $B_y(\mu)$ is an edge-pair-weighted matching objective. Indeed, if $\mu(i) = g$ and $\mu(k) = h$, then the contribution indexed by (i, k, g) is $y_{i,k,g}(v_i(g) - v_i(h))$, which depends on the ordered pair of matching edges (i, g) and (k, h) . CKK show that such 2EBM objectives can be optimized in polynomial time [1]. The following closure observation records the two minor variants of 2EBM that are used below.

Lemma 1 (2EBM closure under additive weights and forbidden edges). *Assume access to the polynomial-time 2EBM oracle of CKK for complete bipartite graphs with arbitrary ordered edge-pair weights. Then, for any edge weights $a_{i,g}$ and any allowed-edge set $E' \subseteq N \times I$, one can optimize*

$$B_y(\mu) + \sum_{i \in N} a_{i,\mu(i)}$$

over all perfect matchings using only edges from E' , or correctly report that no such perfect matching exists, with only polynomial overhead.

Proof. The case $n = 1$ is trivial, so assume $n \geq 2$. First ignore the edge restriction. For every ordered pair of non-adjacent edges (i, g) and (k, h) , define the pair weight

$$\omega_{(i,g),(k,h)} = y_{i,k,g}(v_i(g) - v_i(h)) + \frac{a_{i,g}}{n-1}.$$

For a perfect matching μ , summing these weights over all ordered pairs of distinct edges in μ gives

$$\sum_{i \neq k} \omega_{(i,\mu(i)),(k,\mu(k))} = B_y(\mu) + \sum_{i \in N} a_{i,\mu(i)},$$

because the $k = i$ terms in B_y are zero and each selected edge appears as the first edge in exactly $n - 1$ ordered pairs.

Now impose the allowed-edge set E' . A classical bipartite matching computation first checks whether E' contains a perfect matching. If not, we report infeasibility. If yes, let U be a number larger than the absolute value of the unpenalized 2EBM objective of any perfect matching; for instance, $U = n(n-1) \max |\omega_{e,f}| + 1$ suffices. Replace every pair weight involving at least one forbidden edge by a large negative penalty, say $-3U$. Every matching that uses a forbidden edge then has penalized objective below $-U$, while every allowed perfect matching has objective at least $-U + 1$ under the original weights. Hence an optimal penalized 2EBM solution uses only allowed edges and is optimal among allowed perfect matchings. $\square$

Consequently, for every fixed scalar λ , the problem

$$\max_{\mu \in \mathcal{M}} \left\{ B_y(\mu) + \lambda \sum_{i \in N} \log v_i(\mu(i)) \right\} \tag{6}$$

is solvable by the same oracle whenever the logarithms are well-defined on the allowed edges.

3 A tangent approximation to average Nash welfare

Let

$$\mathcal{M}^+ = \{\mu \in \mathcal{M} : v_i(\mu(i)) > 0 \text{ for every } i \in N\}$$

be the set of positive-welfare matchings. If $\mathcal{M}^+ = \emptyset$, then $\text{aNW}(\mu) = 0$ for every matching μ , and every feasible iEF lottery is optimal. We therefore assume below that $\mathcal{M}^+ \neq \emptyset$.

Let

$$a = \min\{v_i(g) : v_i(g) > 0\}, \quad b = \max\{v_i(g) : v_i(g) > 0\}.$$

For $\mu \in \mathcal{M}^+$ define

$$s_\mu = \frac{1}{n} \sum_{i \in N} \log v_i(\mu(i)).$$

Then $s_\mu \in [\log a, \log b]$ and $\text{aNW}(\mu) = e^{s_\mu}$.

Fix $\varepsilon \in (0, 1)$ and set

$$\eta = \varepsilon/8.$$

Let $T \subseteq [\log a, \log b]$ be a grid containing both endpoints and with consecutive grid points at distance at most η . Thus

$$|T| = O\left(1 + \frac{\log(b/a)}{\varepsilon}\right).$$

For each $t \in T$, let

$$\ell_t(s) = e^t(1 + s - t)$$

be the tangent to e^s at t . Define the approximating welfare coefficient

$$\widetilde{W}(\mu) = \begin{cases} \max\{0, \max_{t \in T} \ell_t(s_\mu)\}, & \mu \in \mathcal{M}^+, \\ 0, & \mu \notin \mathcal{M}^+. \end{cases} \quad (7)$$

Lemma 2 (Pointwise approximation). *For every matching $\mu \in \mathcal{M}$,*

$$(1 - \varepsilon) \text{aNW}(\mu) \leq \widetilde{W}(\mu) \leq \text{aNW}(\mu).$$

Proof. If $\mu \notin \mathcal{M}^+$, then both $\widetilde{W}(\mu)$ and $\text{aNW}(\mu)$ are zero. Assume $\mu \in \mathcal{M}^+$.

Since e^s is convex, every tangent line lies below it. Hence

$$\ell_t(s_\mu) \leq e^{s_\mu} = \text{aNW}(\mu) \quad \text{for every } t \in T.$$

Together with $0 \leq \text{aNW}(\mu)$, this implies

$$\widetilde{W}(\mu) \leq \text{aNW}(\mu).$$

For the lower bound, choose $t \in T$ with $|s_\mu - t| \leq \eta$ and write $d = s_\mu - t$. Then

$$\frac{\ell_t(s_\mu)}{e^{s_\mu}} = e^{-d}(1 + d).$$

For $|d| \leq 1/2$, the elementary estimate

$$e^{-d}(1 + d) \geq 1 - d^2 \quad (8)$$

holds. Indeed, for $f(d) = e^{-d}(1+d)$, we have $f(0) = 1$ and $f'(d) = -de^{-d}$; integrating $|f'|$ from 0 to d gives $1 - f(d) \leq d^2$ on $[-1/2, 1/2]$. Since $|d| \leq \eta = \varepsilon/8 < 1/2$,

$$\ell_t(s_\mu) \geq (1 - \eta^2)e^{s_\mu} \geq (1 - \varepsilon)e^{s_\mu}.$$

Therefore

$$\widetilde{W}(\mu) \geq \ell_t(s_\mu) \geq (1 - \varepsilon) \text{aNW}(\mu).$$

□

4 Solving the approximating LP

Consider the LP (2) with coefficients $c_\mu = \widetilde{W}(\mu)$. We show that its dual admits a polynomial-time separation oracle using polynomially many calls to the 2EBM oracle.

For fixed dual variables y , the separation problem is

$$\max_{\mu \in \mathcal{M}} (B_y(\mu) + \widetilde{W}(\mu)).$$

By the definition of $\widetilde{W}$,

$$\begin{aligned} \max_{\mu \in \mathcal{M}} (B_y(\mu) + \widetilde{W}(\mu)) &= \max \left\{ \max_{\mu \in \mathcal{M}} B_y(\mu), \right. \\ &\quad \left. \max_{t \in T} \max_{\mu \in \mathcal{M}^+} (B_y(\mu) + \ell_t(s_\mu)) \right\}. \end{aligned} \quad (9)$$

The first term is a pure 2EBM objective. For the second term, fix $t \in T$. For $\mu \in \mathcal{M}^+$,

$$\begin{aligned} B_y(\mu) + \ell_t(s_\mu) &= B_y(\mu) + e^t(1-t) + e^t s_\mu \\ &= B_y(\mu) + e^t(1-t) + \frac{e^t}{n} \sum_{i \in N} \log v_i(\mu(i)). \end{aligned}$$

The term $e^t(1-t)$ is constant in μ , and the remaining new term is additive over matching edges. Hence, by Lemma 1, for each fixed t , maximizing

$$B_y(\mu) + \ell_t(s_\mu)$$

over $\mu \in \mathcal{M}^+$ is a 2EBM instance with an additional additive edge weight

$$w_{i,g}^{(t)} = \frac{e^t}{n} \log v_i(g)$$

on positive edges (i, g) , with zero-valued edges forbidden.

Since $|T| = O(1 + \log(b/a)/\varepsilon)$, the separation oracle for the approximating dual uses polynomially many 2EBM calls. The ellipsoid method therefore solves the approximating dual, and the usual dual-to-primal recovery step of CKK returns an optimal primal lottery for the approximating LP with polynomial support.

5 Approximation guarantee

Let Q^* be an iEF lottery maximizing expected average Nash welfare, and let

$$\text{OPT} = \mathbb{E}_{\mu \sim Q^*}[\text{aNW}(\mu)].$$

Let $\tilde{Q}$ be an optimal solution to the approximating LP, i.e., an iEF lottery maximizing $\mathbb{E}[\tilde{W}(\mu)]$.

By Lemma 2,

$$\mathbb{E}_{\mu \sim Q^*}[\tilde{W}(\mu)] \geq (1 - \varepsilon) \mathbb{E}_{\mu \sim Q^*}[\text{aNW}(\mu)] = (1 - \varepsilon) \text{OPT}.$$

By optimality of $\tilde{Q}$ for the approximating LP,

$$\mathbb{E}_{\mu \sim \tilde{Q}}[\tilde{W}(\mu)] \geq \mathbb{E}_{\mu \sim Q^*}[\tilde{W}(\mu)].$$

Again by Lemma 2, $\text{aNW}(\mu) \geq \tilde{W}(\mu)$ for every μ , and therefore

$$\mathbb{E}_{\mu \sim \tilde{Q}}[\text{aNW}(\mu)] \geq \mathbb{E}_{\mu \sim \tilde{Q}}[\tilde{W}(\mu)].$$

Combining the three inequalities gives

$$\mathbb{E}_{\mu \sim \tilde{Q}}[\text{aNW}(\mu)] \geq (1 - \varepsilon) \text{OPT}.$$

We have proved the following theorem.

Theorem 1. *Consider a matching allocation instance with nonnegative valuations and suppose that at least one iEF lottery exists. Let OPT be the maximum expected average Nash welfare over all iEF lotteries. For every $\varepsilon \in (0, 1)$, in the same real-arithmetic and 2EBM-oracle model as the log-Nash algorithm of Caragiannis, Kanellopoulos, and Kyropoulou, one can compute an iEF lottery $\tilde{Q}$ with polynomial support such that*

$$\mathbb{E}_{\mu \sim \tilde{Q}}[\text{aNW}(\mu)] \geq (1 - \varepsilon) \text{OPT}.$$

If $\mathcal{M}^+ \neq \emptyset$, the number of 2EBM calls is polynomial in n and in $1 + \log(b/a)/\varepsilon$, where a and b are respectively the smallest and largest positive valuation in the instance. If $\mathcal{M}^+ = \emptyset$, the average Nash welfare is identically zero and any feasible iEF lottery is optimal.

6 What remains open

The proof above does not give an exact algorithm for maximum expected average Nash welfare. Exact dual separation for that objective asks one to optimize

$$\max_{\mu \in \mathcal{M}} \left\{ B_y(\mu) + \exp \left(\frac{1}{n} \sum_{i \in N} \log v_i(\mu(i)) \right) \right\}.$$

The exponential term is not additive over edges. The tangent construction replaces it by the maximum of finitely many additive lower bounds, which suffices for a multiplicative approximation but not for equality. Thus Theorem 1 should be read as a PTAS/FPTAS-style consequence of the CKK separation framework, not as a resolution of the exact open problem.

Remark 1 (Arithmetic model). The theorem is stated over real arithmetic because the coefficients $\log v_i(g)$ and e^t need not be rational even when the valuations are rational. A Turing-model implementation can approximate these coefficients to polynomially many bits and run the ellipsoid method with certified approximate separation, losing only a further constant factor in ε . This standard numerical issue is separate from the combinatorial reduction to 2EBM.

References

- [1] Ioannis Caragiannis, Panagiotis Kanellopoulos, and Maria Kyropoulou. On interim envy-free allocation lotteries. *Mathematics of Operations Research*, 50(4):3257–3282, 2025. doi:[10.1287/moor.2023.0203](https://doi.org/10.1287/moor.2023.0203). arXiv version: [arXiv:2102.07839](https://arxiv.org/abs/2102.07839).
- [2] Martin Grötschel, László Lovász, and Alexander Schrijver. *Geometric Algorithms and Combinatorial Optimization*. Springer, 1988.
- [3] Alexander Schrijver. *Theory of Linear and Integer Programming*. Wiley, 1986.